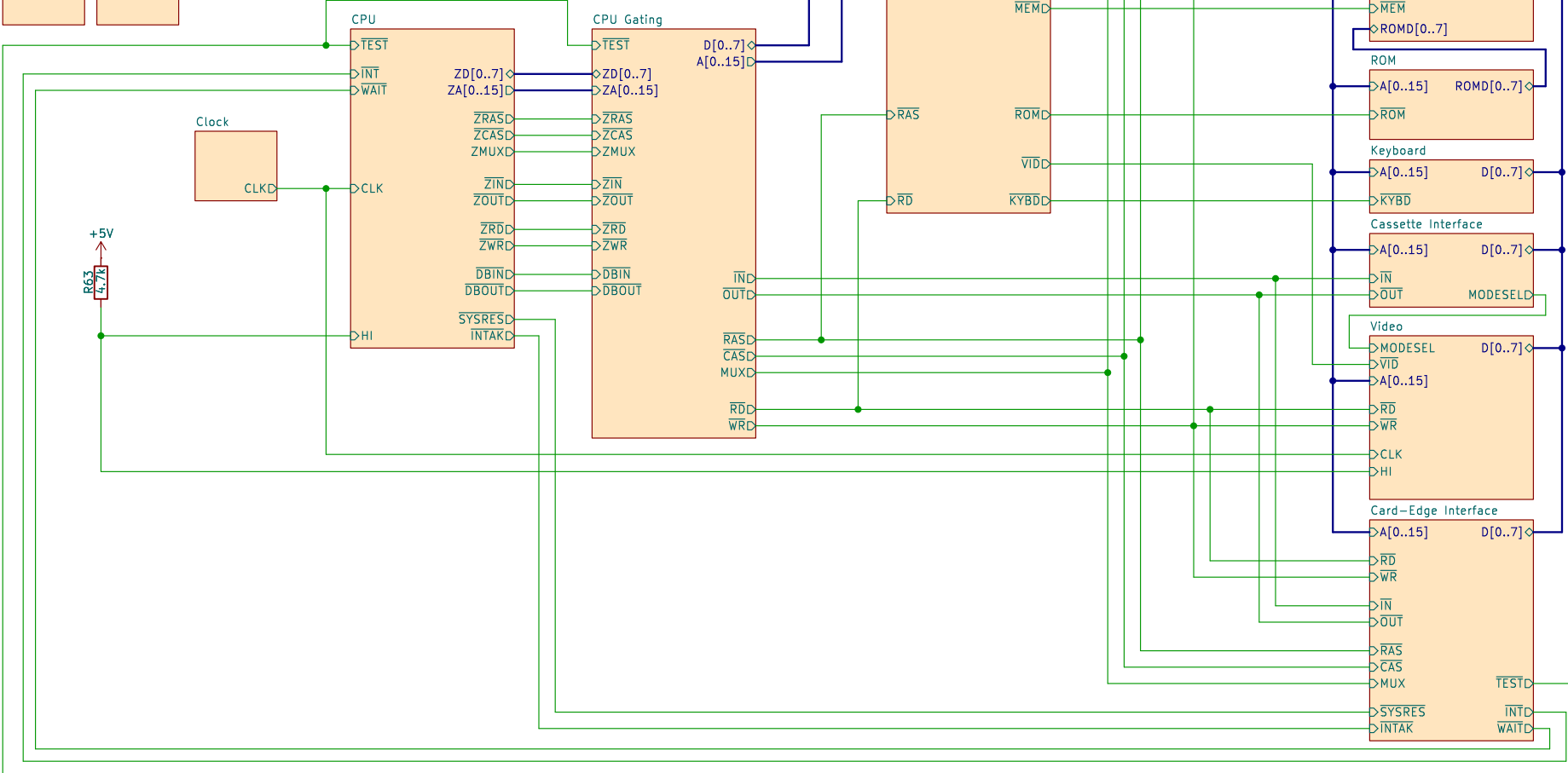
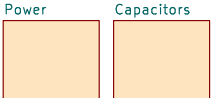


TRS-80 Model I

1700069 E E1



System Overview

Full overview of the system

RetroStack – Marcel Erz

Sheet: /

File: TRS80_ModelI_E1.kicad_sch

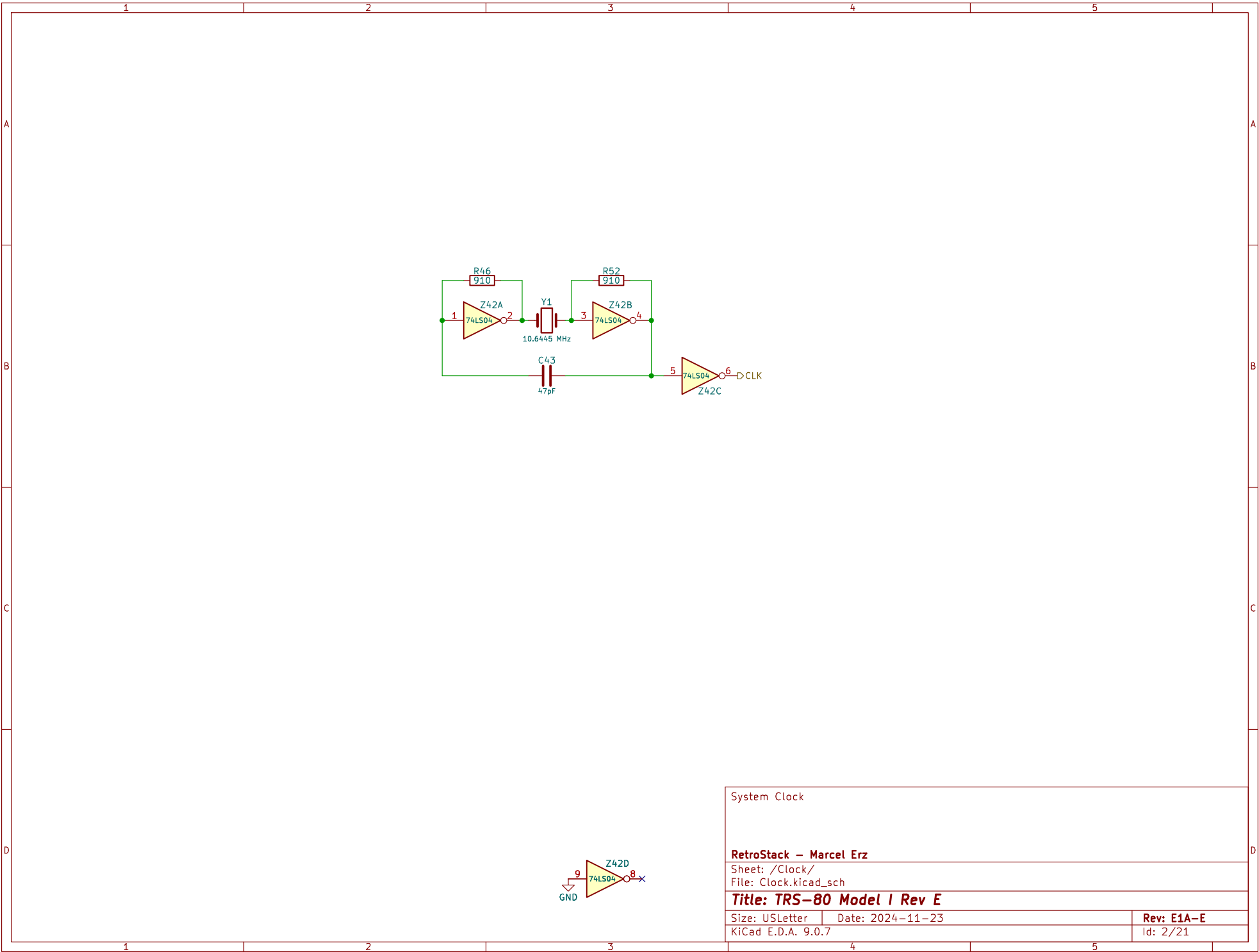
Title: TRS-80 Model I Rev E

Size: USLetter Date: 2024-11-23

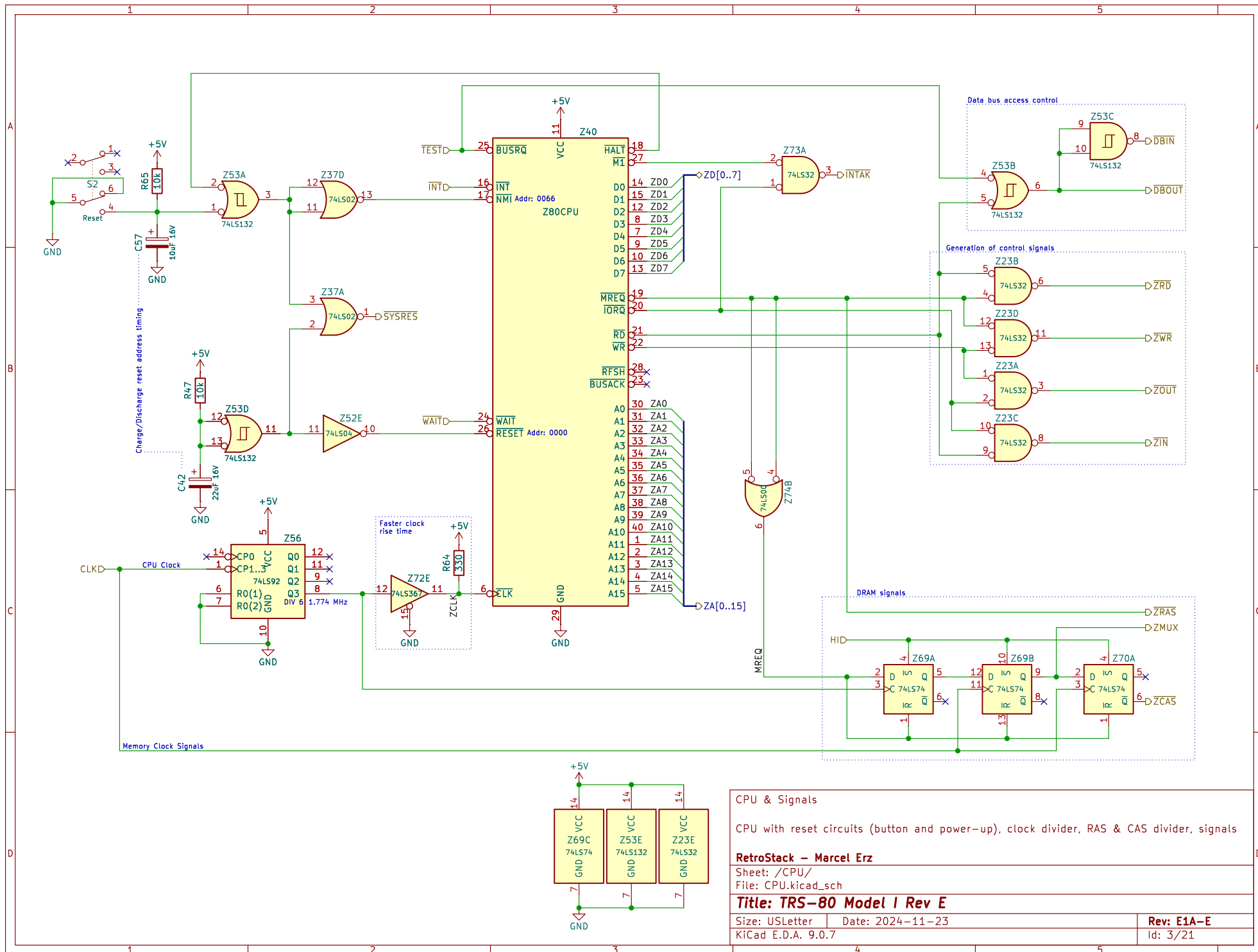
KiCad E.D.A. 9.0.7

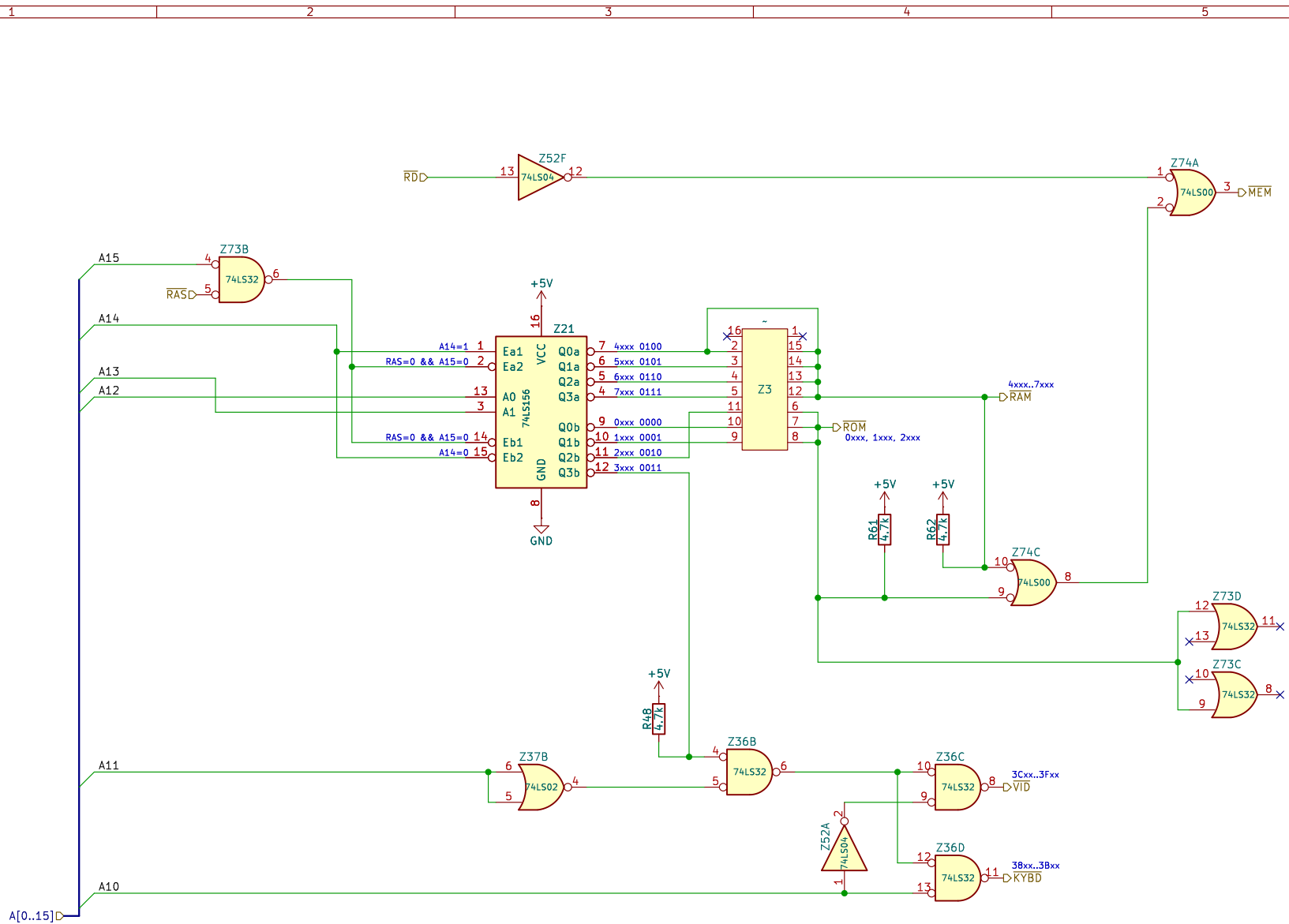
Rev: E1A-E

Id: 1/21



System Clock	
RetroStack – Marcel Erz	
Sheet: /Clock/ File: Clock.kicad_sch	
Title: TRS-80 Model I Rev E	
Size: USLetter	Date: 2024-11-23
KiCad E.D.A. 9.0.7	Rev: E1A-E Id: 2/21





Address Decoder

Address decoder circuit for main board components (except IO)

RetroStack – Marcel Erz

Sheet: /Address Decoder/

File: AddressDecoder.kicad_sch

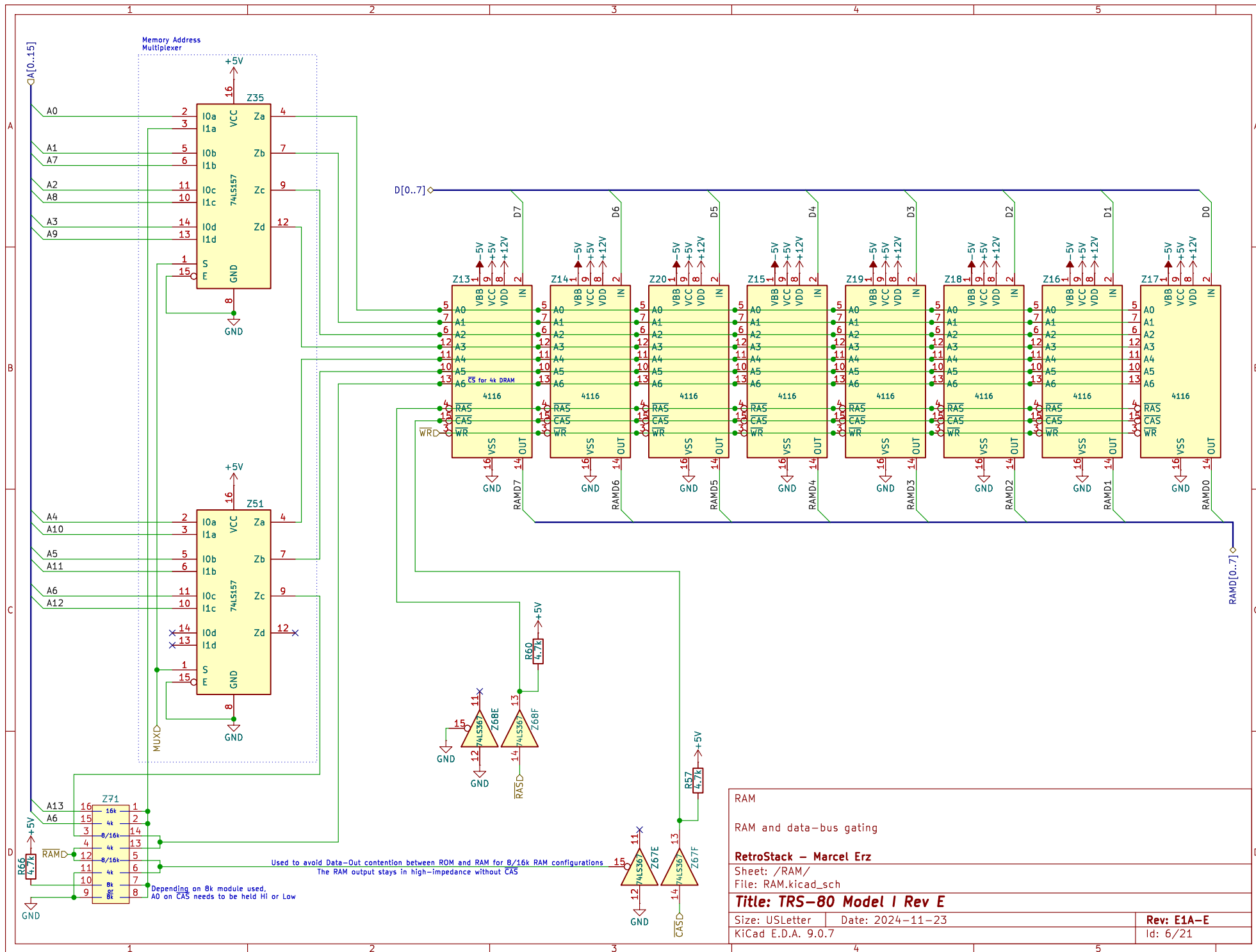
Title: TRS-80 Model I Rev E

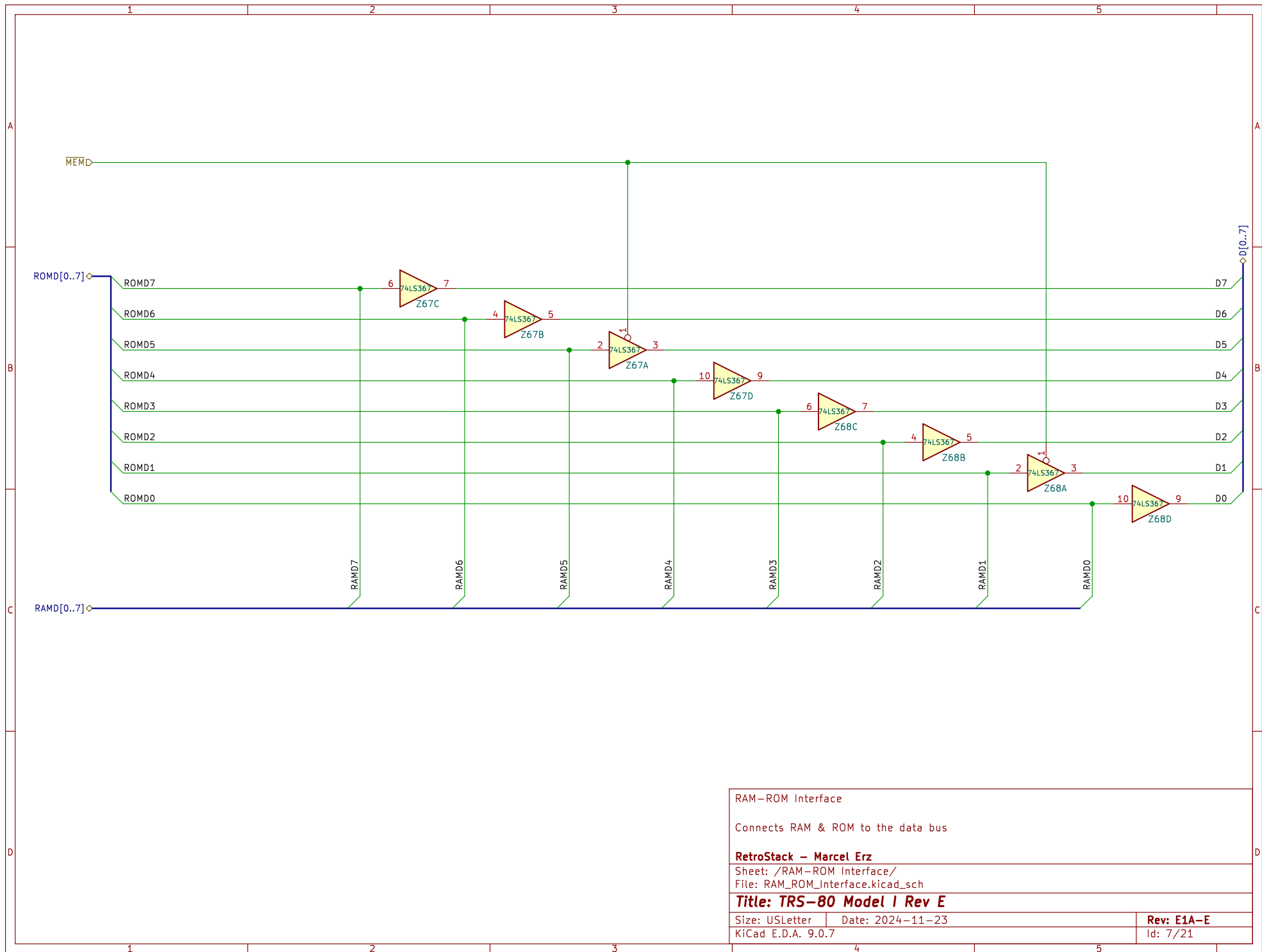
Size: USLetter Date: 2024-11-23

KiCad E.D.A. 9.0.7

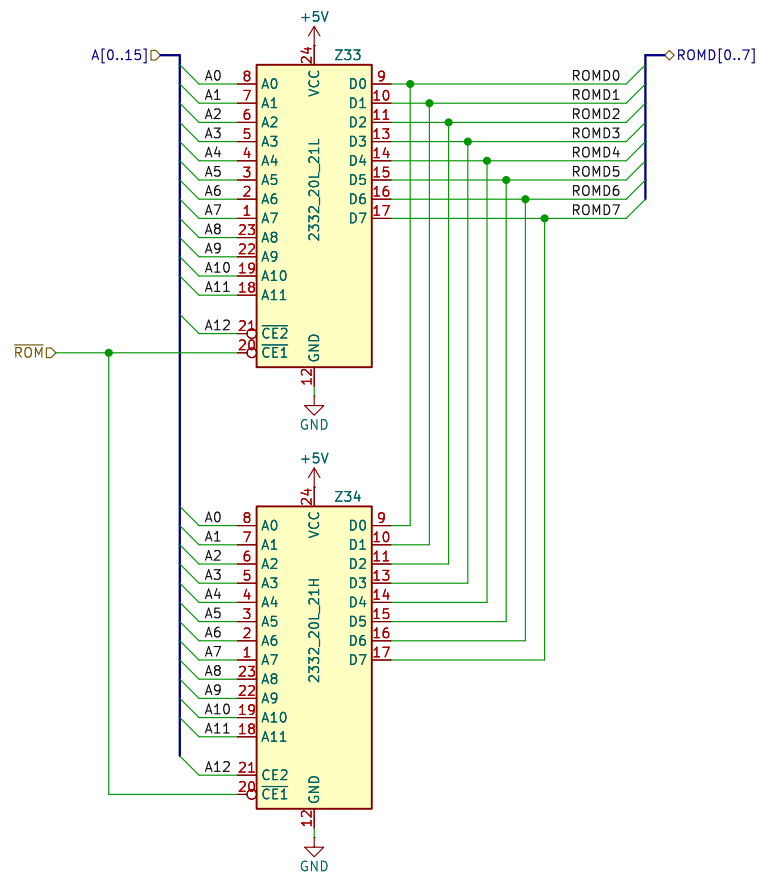
Rev: E1A-E

Id: 5/21





RAM-ROM Interface	
Connects RAM & ROM to the data bus	
RetroStack – Marcel Erz	
Sheet: /RAM-ROM Interface/	
File: RAM_ROM_Interface.kicad_sch	
Title: TRS-80 Model I Rev E	
Size: USLetter	Date: 2024-11-23
KiCad E.D.A. 9.0.7	Rev: E1A-E
	Id: 7/21



- Level I ROMs:
- Intel 2616 EPROM (White Crystal Window)
 - National MM2316 ROM (Colored Ceramic) (R/D=ROM A, S/D=ROM B)
 - Motorola 7800 Series ROMs (7807=ROM A, 7804=ROM B)
 - Motorola 7800 Series ROMs (7809=ROM A & B)
- Level II ROMs:
- Generic 3 ROMs on separate board with logic chip (each 32kbit/4kb)
 - NEC 8043364 & 8043732 (64kbit/8kb & 32kbit/4kb) (early 1979)
 - NEC 8044364 & 8044732 (64kbit/8kb & 32kbit/4kb) (early 1980)

ROM

ROM circuit

RetroStack – Marcel Erz

Sheet: /ROM/

File: ROM.kicad_sch

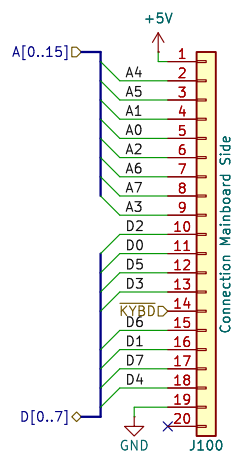
Title: TRS-80 Model I Rev E

Size: USLetter Date: 2024-11-23

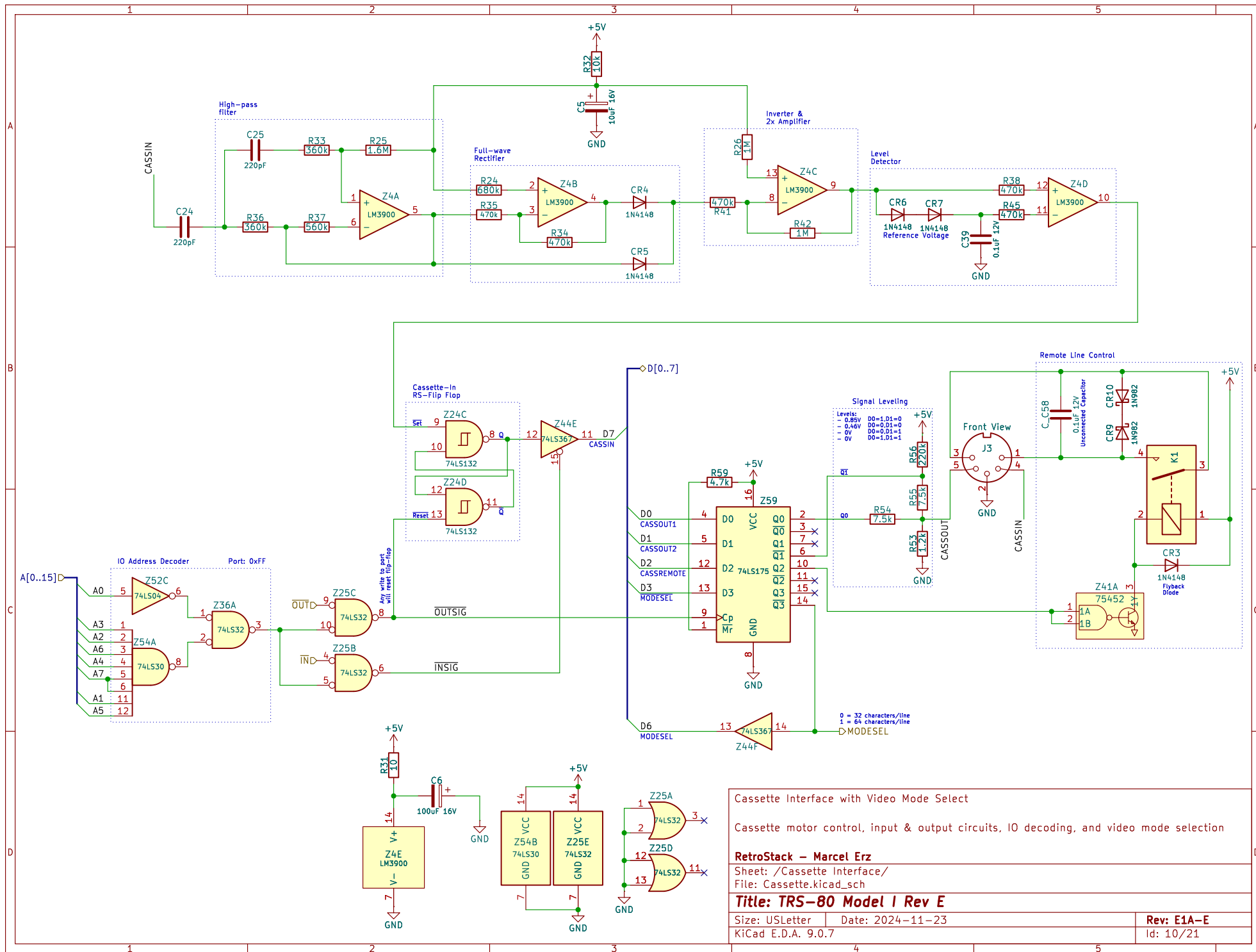
KiCad E.D.A. 9.0.7

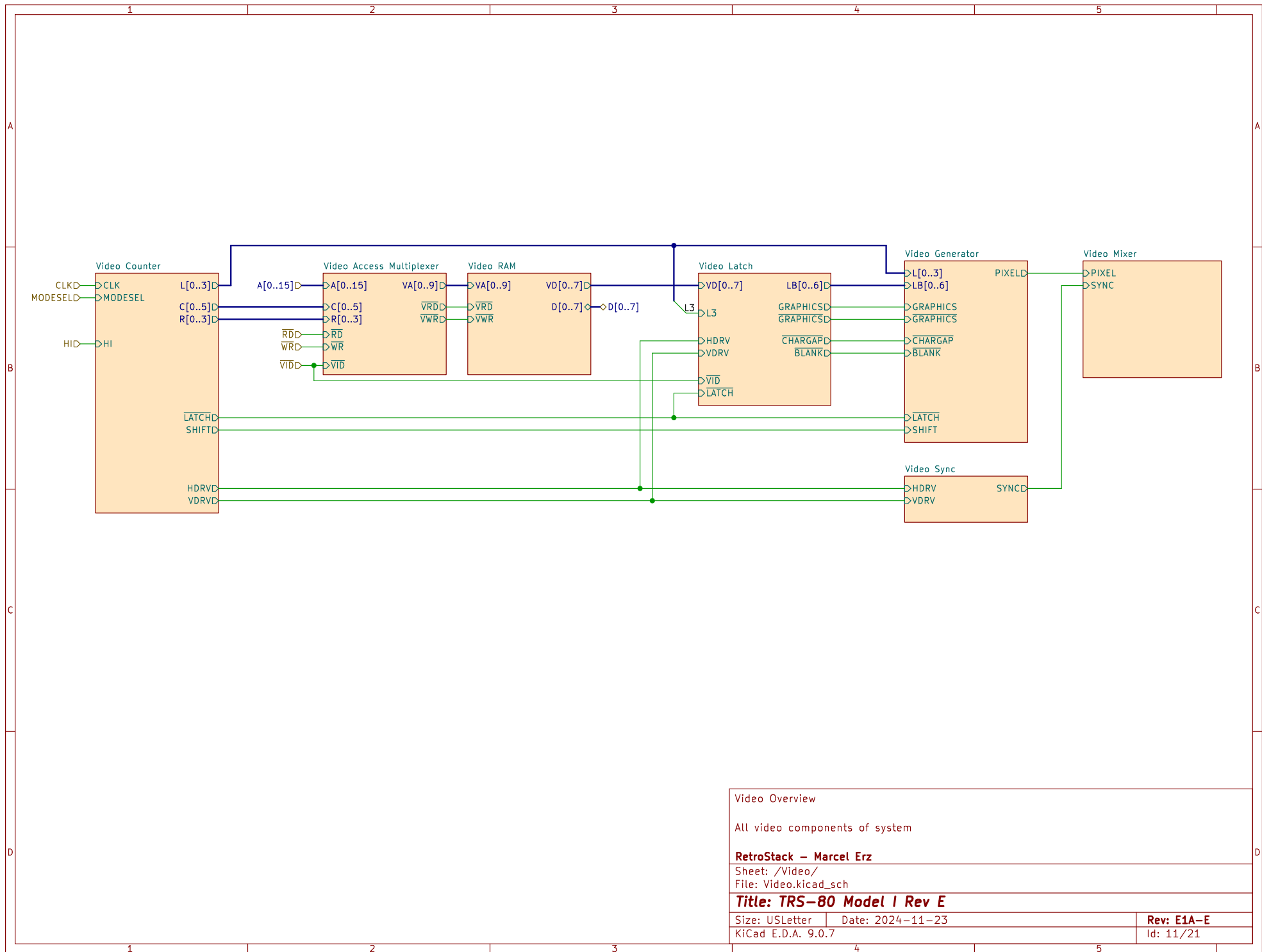
Rev: E1A-E

Id: 8/21

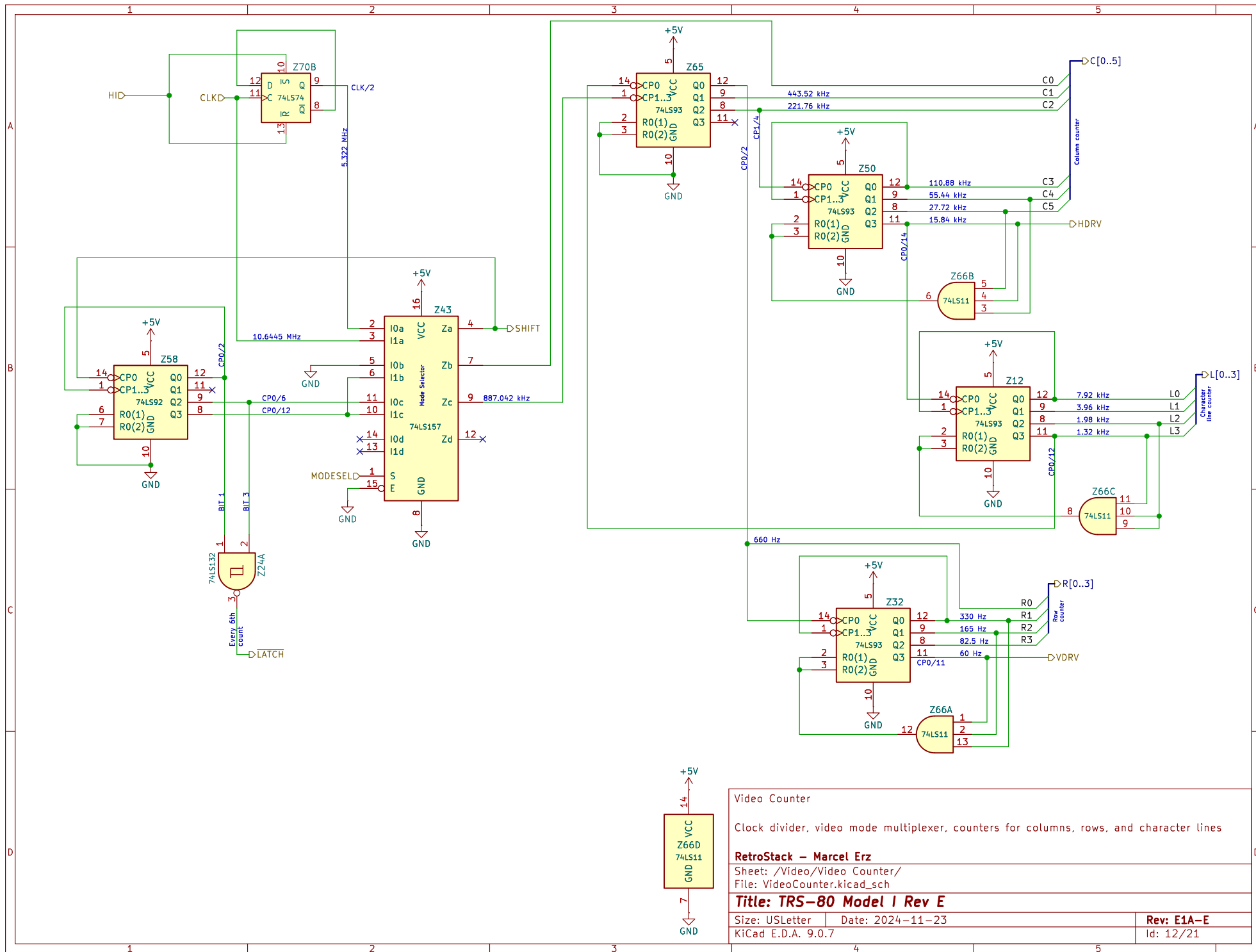


Keyboard & Interface			
Keyboard interface circuit			
RetroStack – Marcel Erz			
Sheet: /Keyboard/ File: Keyboard.kicad_sch			
Title: TRS-80 Model I Rev E			
Size: USLetter	Date: 2024-11-23	Rev: E1A-E	
KiCad E.D.A. 9.0.7		Id: 9/21	





Video Overview		
All video components of system		
RetroStack – Marcel Erz		
Sheet: /Video/ File: Video.kicad_sch		
Title: TRS-80 Model I Rev E		
Size: USLetter	Date: 2024-11-23	Rev: E1A-E
KiCad E.D.A. 9.0.7		Id: 11/21



Video Counter

Clock divider, video mode multiplexer, counters for columns, rows, and character lines

RetroStack – Marcel Erz

Sheet: /Video/Video Counter/

File: VideoCounter.kicad_sch

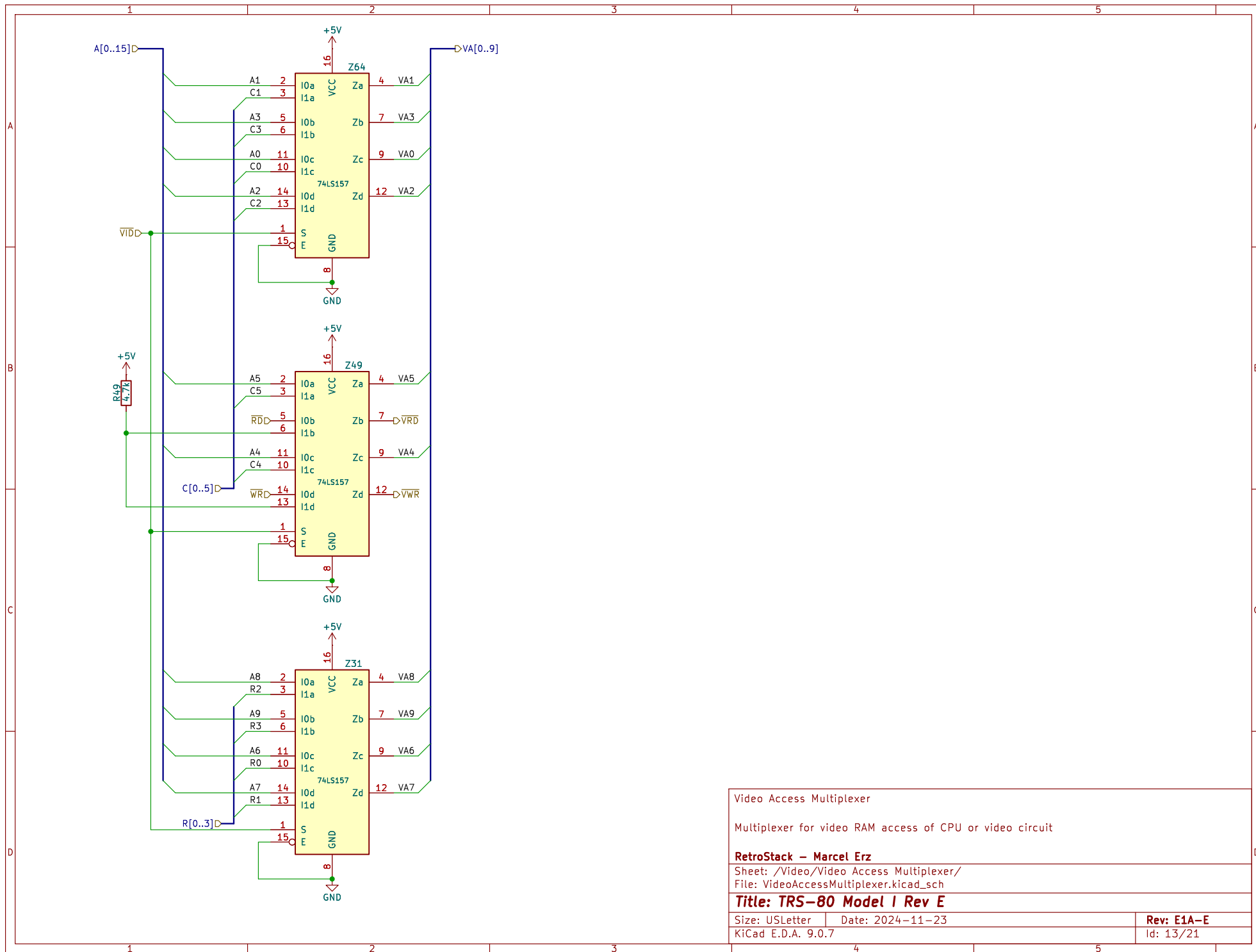
Title: TRS-80 Model I Rev E

Size: USLetter Date: 2024-11-23

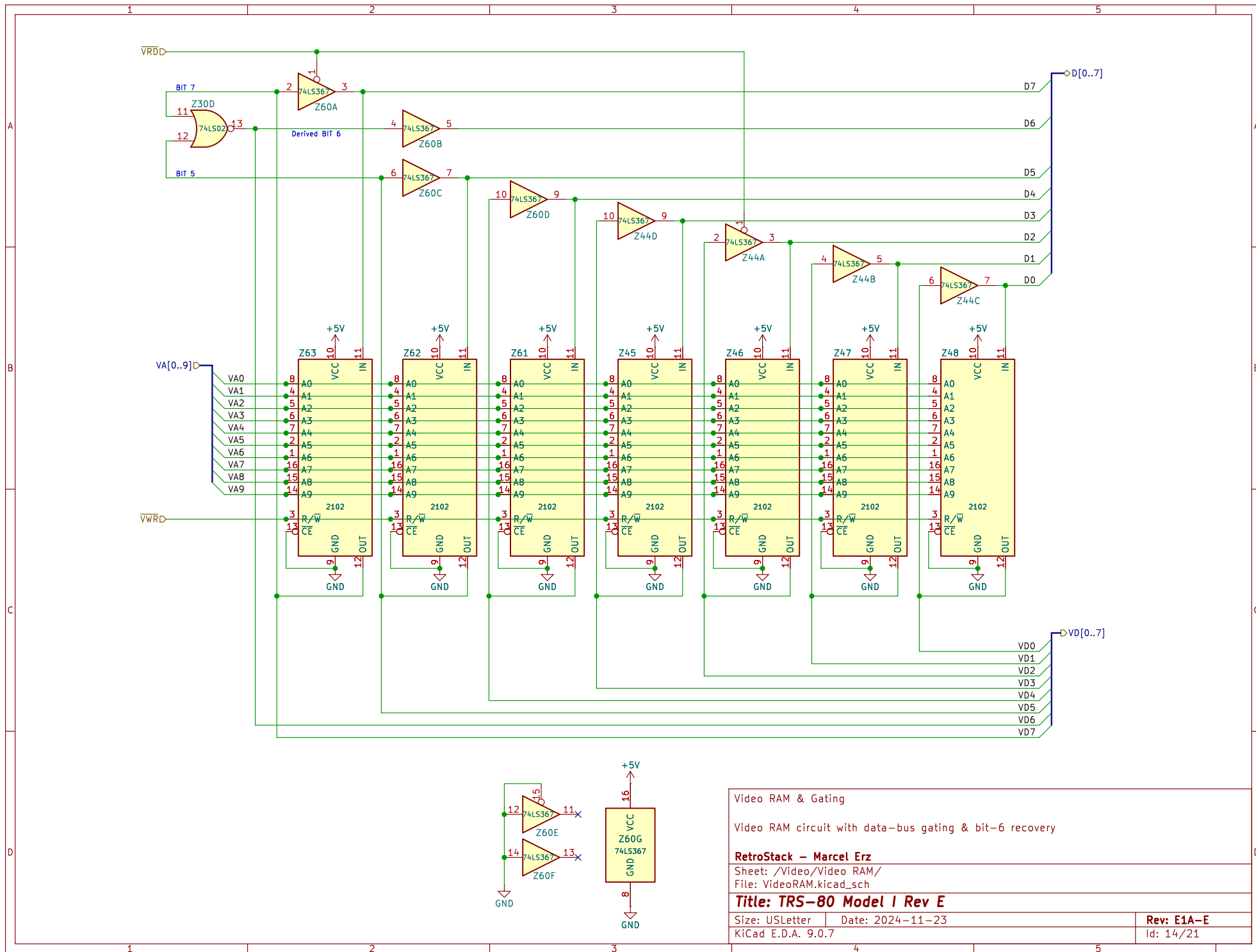
KiCad E.D.A. 9.0.7

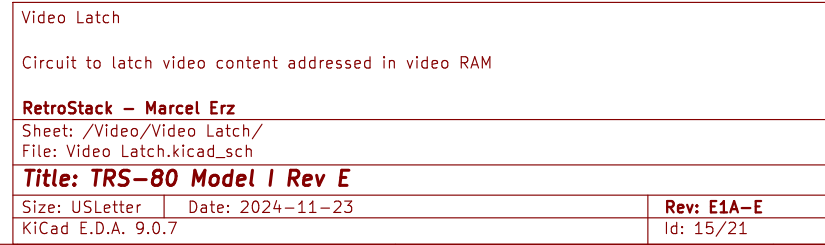
Rev: E1A-E

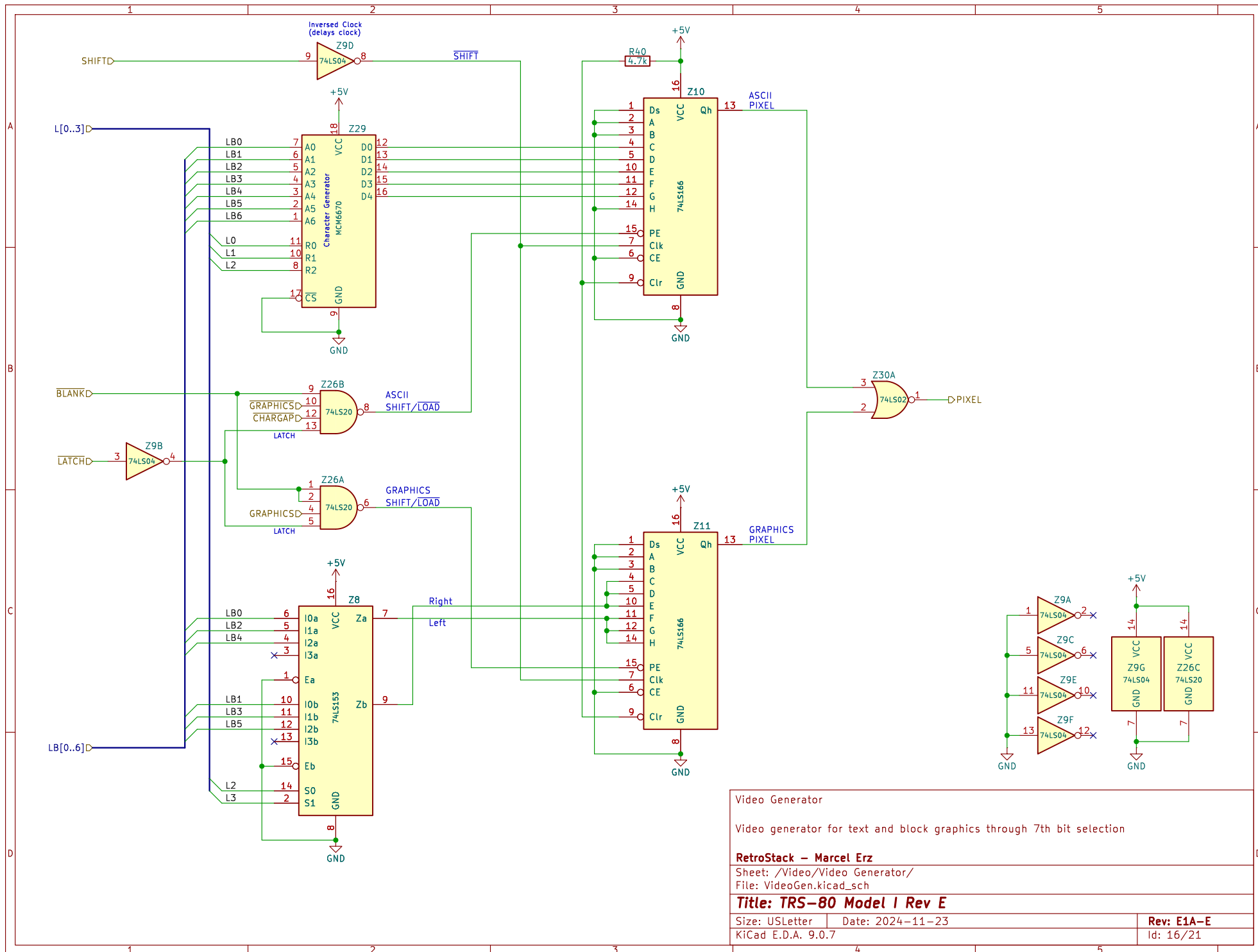
Id: 12/21

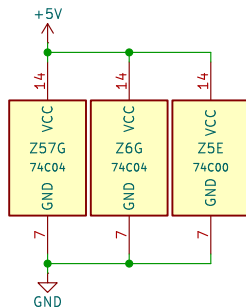
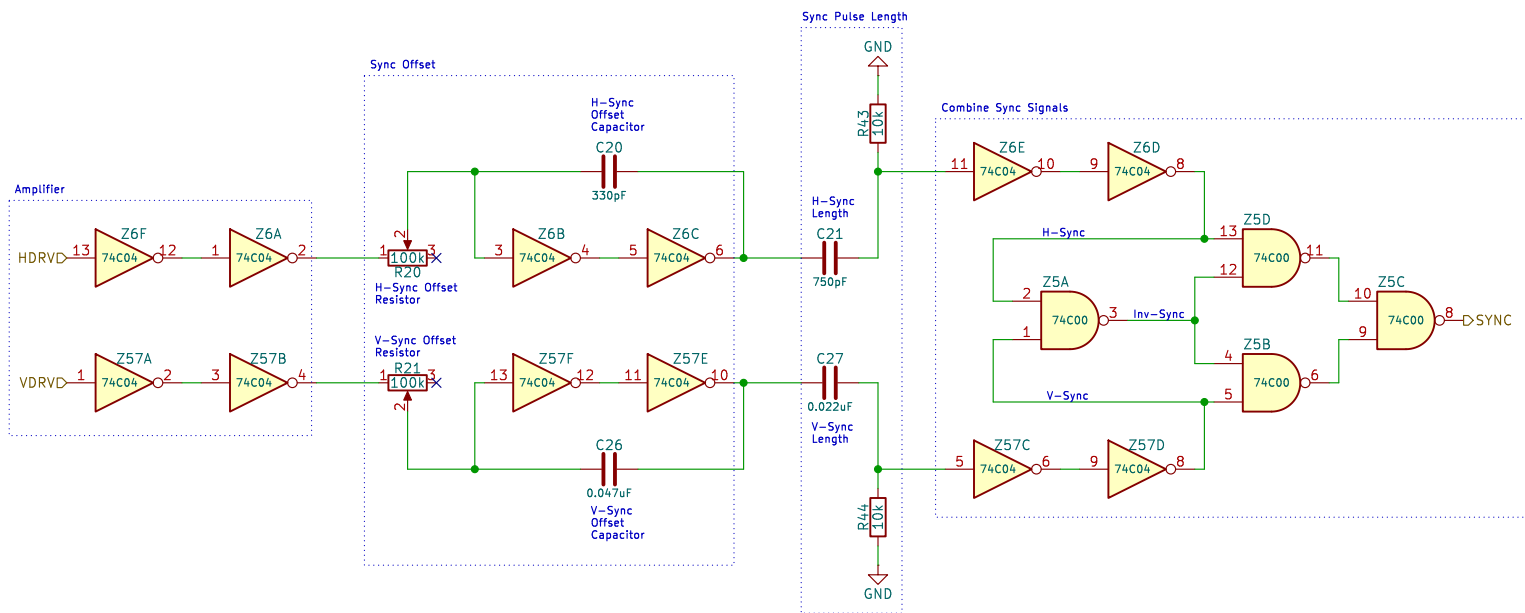


Video Access Multiplexer	
Multiplexer for video RAM access of CPU or video circuit	
RetroStack – Marcel Erz	
Sheet: /Video/Video Access Multiplexer/ File: VideoAccessMultiplexer.kicad_sch	
Title: TRS-80 Model I Rev E	
Size: USLetter	Date: 2024-11-23
KiCad E.D.A. 9.0.7	Rev: E1A-E Id: 13/21









Video Sync

Circuit to combine sync signals and defining front- and back-porches in video signal

RetroStack – Marcel Erz

Sheet: /Video/Video Sync/

File: VideoSync.kicad_sch

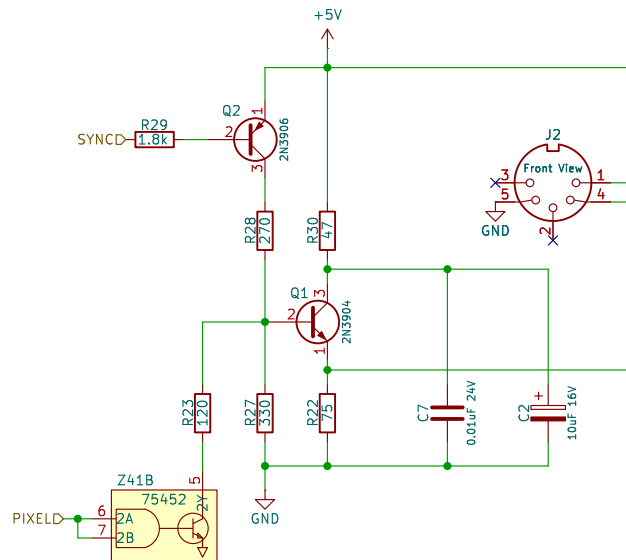
Title: TRS-80 Model I Rev E

Size: USLetter Date: 2024-11-23

KiCad E.D.A. 9.0.7

Rev: E1A-E

Id: 17/21



Q2 open, Z41 any => 0V (sync level)
 Q2 closed, Z41 closed => 1.23V (black level) – R28 with series resistors R23 & R27
 Q2 closed, Z41 open => 2.75V (white level) – Only R27 & R28

Video Mixer	
Mixing circuit for sync and pixel data	
RetroStack – Marcel Erz	
Sheet: /Video/Video Mixer/ File: VideoMixer.kicad_sch	
Title: TRS-80 Model I Rev E	
Size: USLetter	Date: 2024-11-23
KiCad E.D.A. 9.0.7	Rev: E1A-E Id: 18/21

